Aptiv 2021 Technology Kickoff Event: Innovation in Motion

January 11, 2021 | Virtual Event Hosted by Aptiv Investor Relations

WELCOME

OPENING REMARKS*: Elena Rosman | Vice President Investor Relations*

Welcome and thank you for joining Innovation in Motion, Aptiv's 2021 Technology Kickoff event. On behalf of the Aptiv team, we are truly excited to start 2021 with you and we hope you are all staying healthy and safe. Over the last year, our teams have stayed focused, creating technology solutions that help address some of the industry's toughest challenges, while pushing the envelope to bring solutions to market quickly.

While we are not able to be together in person this year, we are truly excited to kick off the year with this virtual event - beginning with a video presentation showcasing some of the latest technologies in discussions with customers; followed by a live 30 minute Q&A session with myself and Glen De Vos, Aptiv's Chief Technology Officer, answering any questions you may have. To ask a live question, please use the conference dial-in provided on this event site, featured at the bottom of your screens. the line is now open and you may enter the queue by pressing *1.

Before we get started, I wanted to remind you that today's presentations include forward looking statements, based on our current view of future financial performance and may be materially different from our actual performance for reasons we cite in our form 10K and other SEC filings. We will not be addressing questions about the quarter our guidance during today's event, and will defer any financial questions until our Q4 2020 earnings conference call in a couple weeks.

Now that we have the housekeeping taken care of, today's program will feature: a number of our technology leaders showcasing groundbreaking Aptiv innovations -- advancing the development and deployment electrified, software-defined vehicles of the future. With that, let's get started!

INNOVATION IN MOTION VIDEO

WELCOME MESSAGE*: Kevin Clark | President and Chief Executive Officer*

Hello everyone, and welcome to Aptiv's 2021 Kickoff Event celebrating Innovation in Motion! We are excited to share the progress we've made over the past year, leveraging our unique industry position as the only provider of both the Brain and Nervous System of the vehicle, making mobility safer, greener and more connected.

Never has Aptiv's mission of delivering sustainable solutions had more meaning for our society than it does today. The pandemic has led many of us to re-evaluate how we view safety, efficiency, comfort and convenience.

Throughout the pandemic, Aptiv kept our most important asset – our employees – safe, and we were able to continue innovating and delivering solutions for our customers. Throughout today's event, you will see firsthand how we have kept Innovation in Motion.

As the economic recovery takes hold, Aptiv is enabling our customers to thrive – accelerating the transition to an electrified, software-defined vehicle – by employing a more holistic engineering and development approach to optimize the software and system solutions that span the full vehicle stack.

We solve the industry's toughest challenges, creating exceptional value for customers, which in turn, generates best-in-class returns for shareholders.

Now there's a lot to talk about. So let's get started. Let me turn it over to Glen De Vos, our Chief Technology Officer., who will lead today's discussion around our expanded capabilities in delivering full stack solutions – all of which are on the path to Smart Vehicle Architecture.

ENABLING THE SOFTWARE DEFINED VEHICLE WITH SMART VEHICLE ARCHITECTURE™ (SVA™)

Glen De Vos | Senior Vice President and Chief Technology Officer

Thanks, Kevin. While I miss being together in Las Vegas at the Aptiv Pavilion, I am just as excited to showcase our Innovation in Motion! We're all here today – virtually – because our industry is at an inflection point.

We continue to see an acceleration of powertrain electrification – driven by more stringent CO2 regulations and increasing demand globally. Consumers are also demanding more safety and connectivity features, which are increasingly delivered through software; and this increase in software brings with it more advanced sensing and compute platforms to power it. All of this has put tremendous strain on the classical approach to vehicle architectures, and that's where we find ourselves today…Solving these challenges has the potential to both meaningfully improve and save lives.

Our ability to leverage our unique full-stack systems capabilities is helping our customers realize their future technology roadmaps. They understand that changes in vehicle architecture are critical to deliver the feature rich, highly automated vehicles they need in the future; as a result, our customers are converging around new architectures to deliver the higher-contented, more efficient vehicles of the future, where we know how to provide value.

We first introduced the vehicle solution stack to you at CES 2019, and we've continued to showcase our value-add across the stack – from sensor to cloud – thanks to our unique position as the only provider of both the brain and the nervous system.

As a result, we serve as a strong collaboration partner for increasingly complex architectures on the path to SVA™, with the industry-leading capabilities in power and data, compute, perception systems, software, and sensor fusion – which you will hear more about throughout today's presentation.

Our customers recognize Aptiv as a key technology partner, capable of both the design and manufacturing of advanced hardware, fully-integrated into the most complex vehicle systems -- which combined with our differentiated and modularized software capabilities creates value for our customers at every level of the stack -- accelerating their development of the safe, green and connected features consumers want, with the proven automotive-grade systems they can trust. As a result, we are helping our customers democratize technologies as they mature – delivering higher value, at lower cost, for more people.

Smart Vehicle Architecture, or SVA, represents Aptiv's vision for the full electrical and electronic architecture of the vehicle. It includes all of the elements we believe are necessary to simplify the feature-rich and highly complex vehicles of today while efficiently enabling the electrified, software-defined and highly automated vehicles of tomorrow.

This scalable approach also lowers the total cost of ownership for the OEM, while also unlocking the opportunity for Aptiv to capture more value in the vehicle. It is important to be at the point of aggregation, creating new hardware and software revenue opportunities for Aptiv.

Here to share our vision for SVA in more detail is Lee Bauer.

Lee Bauer | Vice President, Advanced Engineering - Smart Vehicle Architecture

Thanks, Glen. Since we unveiled SVA last year, Aptiv has made key advancements in high voltage electrification, domain compute and zone control solutions – all of which are necessary steps along the path toward fulfilling the SVA vision.

But let me start by reiterating some the key elements of our approach. SVA leverages Aptiv's unique position with both the brain and the nervous system of the vehicle, and our equally unique understanding of the challenges caused by the complexity of today's vehicle architectures.

Our approach delivers value to our customers by helping them balance the performance and cost trade-offs they face by managing how content and value are allocated around the vehicle by adhering to three fundamental principles: we abstract (or separate) hardware from software; we separate Input/Output from compute and up-integrate multiple single-purpose ECUs; While each of those steps has its own benefits, when taken together we have the ability to "serverize" compute – resulting in a truly software-defined architecture that's sustainable well into the future.

That's SVA in a nutshell. More specifically…SVA provides efficient, affordable and safe scalability of new features and functions that customers want and more importantly that customers are willing to pay for. It streamlines and accelerates the development process by separating hardware and software development into two independent and parallel paths which also significantly increasing reuse. It creates logical and modular subassemblies, simplifying manufacturing and assembly, decreasing complexity and costs while improving quality. Finally, open platforms on central compute with full OTA capability create a new and vibrant third-party ecosystem of apps for monetization that were not previously possible in automotive.

I like to describe it as "the sum of all our parts", and so as you would expect, the insights and innovations we talk about today typically leverage multiple parts of that architecture vision to help solve some of the industry's toughest challenges.

Glen De Vos | Senior Vice President and Chief Technology Officer

Thanks Lee. SVA is all about value. By design, it will effectively lower the total cost of ownership from development through manufacturing and then into the field. Faster development and software lifecycles will enable new functionality and business models and allow our OEM customers to clearly differentiate.

But, SVA also provides tremendous value for Aptiv. First, it increases our addressable content in both hardware and software value, strengthening our competitive position. Second, it brings efficiency through automation and improved reuse. And finally, it unlocks new functionality that enables new and accretive business models for us as well as our OEM customers who increasingly want to be in control of the software that defines the user's experience.

There has historically been more momentum with the luxury OEMs, but more recently we have seen a lot of interest with those who also operate in the mass market. Our solution scales the total market by managing software and wire harness complexity and lowers the associated cost of both – while also unlocking new content opportunities for Aptiv.

This represents a meaningful add to our existing portfolio, enabled by the abstraction of hardware from software, as well as our ability to monetize the SVA platform, which includes a comprehensive development and testing environment needed for these new applications that will be up integrated on our Centralized Control.

As I said earlier, it is important to be at the point of aggregation, and new hardware, software and more sophisticated engineered components drive higher value revenue opportunities for Aptiv. Our favorable position in domain controllers and expertise in active safety provides a terrific launching pad as we work with our customers on their next-generation architecture solutions.

UNVEILING OF APTIV'S NEXT GENERATION ADAS PLATFORM FOR SAFE MOBILITY

Glen De Vos | Senior Vice President and Chief Technology Officer

As we think about the impact on vehicle architectures, there are few - if any - that have been more disruptive from a data standpoint than ADAS. The proliferation of safety critical sensor data puts significant challenges on reliable signal distribution.

Many of us in the industry share a common goal: a future with zero traffic accidents and zero fatalities. While this is undeniably ambitious, we are maximizing the benefits of the advanced safety systems we are deploying, while working hard to reduce the integration costs associated with deployment of these systems, which have the opportunity to save millions of lives.

According to the World Health Organization, the lives of about 1.3 million people are cut short as a result of road traffic accidents every year. More than half of those deaths involve vulnerable road users: pedestrians, cyclists and motorcyclists. Another 20 to 50 million others suffer non-fatal but nonetheless life-changing injuries. And road traffic collisions cost most countries an estimated 3% of their gross domestic product. However, the good news is that most accidents are preventable. According to NHTSA, 94% of traffic accidents are caused by human error.

At Aptiv, we believe in imagining a better future – but we also believe in making it real and affordable. That means delivering technologies that are flexible and scalable, that help protect vulnerable road users, and that democratize safety technology so that it is within reach of all drivers, everywhere.

To address those needs, Aptiv is proud to unveil Aptiv's Next Generation ADAS platform for safe mobility. This platform combines our best-in-class sensing and perception solutions with a scalable approach, an open development environment, and lifecycle maintenance and enhancement capabilities.

Aptiv leverages our deep systems expertise and learnings from deploying the industry's largest, most diverse safety installed base over the last 20 years, with active safety technologies in use by 20 different global automakers.

Aptiv builds on the strong foundation of Satellite Architecture, which is being deployed today by multiple OEMs on vehicles around the world and is expected to be installed on 10 million vehicles by 2025.

We believe the platform provides the tools to solve the industry's toughest challenges around safety. At the same time, it can reduce up-front development and integration costs by supporting the pre-integrated features "out of the box."

Now let's take a closer look at the elements that make up Aptiv's next generation ADAS. Creating the best advanced safety platform starts with the most comprehensive and reliable environmental model. Here to introduce the Aptiv Environmental Model is Christian Nunn.

ENVIRONMENTAL MODEL

Christian Nunn | Global Advanced Chief Engineer

Thank you, Glen. Our portfolio of external and internal sensing solutions is fully integrated across multiple sensing modalities and capable of 360 degree coverage. This provides a rich data set to create the next generation environmental model, which fuses data from multiple sensors and sensor modalities together while applying AI and Machine Learning techniques to better interpret the data.

That enables us to get more information out of the data, so that the system can accurately and reliably identify objects around a vehicle. As a result, we're able to reduce the total perception system costs, while maintaining and expanding the conditions under which the vehicle can safely and effectively operate.

Aptiv is using machine learning strategically. Instead of taking a brute-force approach and applying machine learning to all of the raw data provided by the system, let's say, a radar, we first perform classical preprocessing and then apply machine learning to just those areas where it makes sense. Without this interim step, an AI system would have to be extremely powerful, and more expensive and more resource-intensive. It would require long training sequences and would be difficult to troubleshoot. Aptiv's approach is much more efficient.

However, even the best fusion and environmental models rely on the sensing inputs which feed them. That's why Aptiv continues to push the envelope on sensor technology. Here to give us the latest is Gabor.

SENSORS

Gabor Vinci | Global Product Line Manager

Thanks, Christian. Aptiv incorporates our full line of sensors, including advanced automotive radars. Today we are announcing two families of next-generation radars, the SRR6 and the FLR4.

The SRR6 family of corner radars offers up to twice the detection range versus the previous generation while improving angular resolution by as much as 3x.

The FLR4 family is our next generation of forward-facing radars, providing twice the range resolution and three times the vertical field of view of previous radars. The FLR4+ is Aptiv's first truly 4D imaging radar, capable of meeting the most demanding applications at an attractive price/performance ratio. It improves range detection by over 60% over previous generations, and its elevation target discrimination, combined with machine learning capabilities in signal processing, allow vehicles to figure out at long distance and high speed whether they can drive over an object.

Both of these sensor families have the flexibility to support either satellite or smart sensor configurations. Ensuring these sensors scale in a flexible and cost efficient manner is key. With that in mind, let me turn it over to Parham to discuss how Aptiv is taking scalability further than ever before.

ADDITIVE SCALABILITY

Parham Vasaiely | Global Director, Product Management Advanced Safety and Autonomous Driving

Thanks, Gabor. An important aspect of Aptiv's ADAS platform is what we call additive scalability. With additive scalability – as with our Satellite Architecture – each configuration builds on the previous one, which reduces your design and engineering costs, simplifies the interface into the vehicle electrical architecture, and improves lifecycle management.

Aptiv has specified a sequence of "reference" software and hardware configurations, which are highly configurable by each customer, and range from entry-level safety compliance through level 3 premium and luxury performance.

OPEN PLATFORM

Glen De Vos | Senior Vice President and Chief Technology Officer

Thanks, Parham. That kind of flexibility extends to the way we develop software for the platform. Aptiv continues that acceleration toward software-defined platforms.

Aptiv's ADAS platform allows developers, Aptiv, and other third parties to easily innovate on top of it. By adopting business models that support creation of advanced features, and by encouraging an ecosystem of partners, we're able to drive greater innovation and unlock value

Our ADAS platform also fits perfectly into our SVA, with its centralized approach and fundamental building blocks for creating software-defined vehicles.

Now these vehicles are not going to stop progressing once they roll off the assembly line. So here to tell us more about lifecycle management and applying a software defined approach to advanced safety is Jada Smith.

CONNECTION TO FUTURE ARCHITECTURES

Jada Smith, Global Engineering Director, Software Platforms

Thanks, Glen. Aptiv enables our customers to move towards the software-defined platforms envisioned by SVA™ in several important ways. It abstracts hardware from software with well-designed, standardized interfaces for sensors and feature functions. It provides a common software integration platform capable of supporting significant development reuse, and leveraging both classic AUTOSAR and AUTOSAR Adaptive standards. And it separates I/O from compute for OEMs pursuing zonal concentration of I/O.

These are the design principles that will take us into a future of highly automated vehicles. With over a decade of experience, we know what it takes to support affordable, fail-operational performance for power distribution, network stability and compute availability and performance.

From the first coast-to-coast automated drive in 2015, to the driverless testing we have ongoing in our Motional joint venture with Hyundai, these experiences allow us to better understand the challenges you will face, regardless of timelines or technology roadmaps. And speaking of timelines…

LIFECYCLE MAINTENANCE AND ENHANCEMENT

Jada Smith, Global Engineering Director, Software Platforms

We've designed Aptiv's ADAS platform to evolve and improve over time, while minimizing revalidation and deployment costs. It supports over-the-air, or OTA, updates and enhancements over the program life, providing a scalable, low-risk and cost-effective way to improve the user experience long after vehicles leave the factory, and we are enabling new business models in the process.

Our Continuous Integration and Continuous Deployment tooling allows our customers to quickly develop and securely deliver solutions, enhancing their ability to differentiate themselves with their consumers. Our open platform is designed to be able to easily test, integrate and deploy enhancements as they happen, regardless of where that innovation occurs. And by centralizing compute power within a vehicle, Aptiv greatly simplifies security and OTA management.

Glen De Vos | Senior Vice President and Chief Technology Officer

Thanks, Jada. Aptiv provides all the tools you need to create best-in-class active safety systems that scale. As we mentioned, one of the key elements in Smart Vehicle Architecture that enables this kind of innovation is the separation of I/O from compute. We achieve that by implementing Zone Control. Martin Bornemann is here to tell us more.

ACCELERATING CENTRALIZATION WITH APTIV ZONE CONTROL

Martin Bornemann | Director, Systems and Architecture

Thanks, Glen. A few strategically placed zone controllers can significantly reduce complexity and cost today, while building the right architectural foundation for the vehicles of tomorrow.

As you move from distributed architectures to domain control, in order to support advanced, compute-intensive functionality such as active safety and user experience, zonal implementations are highly complementary. Together, they bring us a host of benefits to today's architectures, which Aptiv has validated across multiple OEMs during the past year.

With our unique perspective of both the brain and the nervous system of the vehicle, Aptiv is able to take a holistic, systems approach to zone and domain control, balancing tradeoffs in managing power, data and compute, making Aptiv ideal partner for this evolutionary step. Here to take us through the benefits of zone control is Christian Schaeffer.

Christian Schaefer | Director, Electrical and Electronic Systems

Thank you, Martin. Zone control has several benefits…

Zone controllers can act as an up-integration point for multiple single-purpose ECUs. Up-integration reduces physical complexity by eliminating individual boxes and all of the redundant power supplies, housings and connectors that go with them. In one study for an OEM, Aptiv found it could remove over 8 kilograms of weight, while significantly reducing cost and packaging space.

Because zone controllers handle all of the physical and logical connections to peripheral sensors and devices, the architecture allows for a common, optimized power and data backbone from the zone controller to the domain controllers.

Zone controllers are a less expensive upgrade point as the need for additional I/O expands over time. For example, if you were to add another radar, you would only have to swap in a new zone controller to physically connect to it, while leaving the more expensive domain controller in place and avoiding the additional certification that would go with it.

The architecture wiring harnesses, reduces labor costs and enables automation of assembly. An Aptiv study found that zone control could enable about 60% automation for one of our customers, although this is dependent on component design implementation.

As power distribution hubs, zone controllers become the logical location for smart fusing. This is a key area where Aptiv's brain and nervous system capabilities have helped to optimize performance, so let's take a moment to look at this in more detail.

Aptiv replaces traditional melting fuses in relays with semiconductors, resulting in a number of advantages: The first is savings in power distribution cabling. In the past, wires needed to allow enough tolerance for peak load without having a melting fuse blow. By contrast, wires can be specified to the physical limit of the load over a specified period of time. That often means a reduction of wire gauge and therefore a reduction in weight and cost. The second advantage is better energy management. This is especially critical with electric vehicles, where if the battery is running low, the system can use smart fuses to switch off functions judiciously, for brief periods of time throughout the vehicle. A third advantage is in reliability and predictive maintenance.

With Aptiv's unique portfolio, we are perfectly positioned to bring this solution to market.

Lee Bauer | Vice President, Advanced Engineering - Smart Vehicle Architecture

Thanks Christian, clearly a significant benefit from zone control. To help our customers advance their zone control strategy, Aptiv leveraged insights from our unique position with both the brain and nervous system of the vehicle to define multiple variants. Each are optimized for cost and performance requirements supporting battery electric platforms, different vehicle types, up-integration strategies, and levels of automation on the path towards the software defined vehicle.

All variants include Safe power distribution and load management delivering reliable power to the sensors, peripherals and actuators which require it. With integrated smart fusing and intelligent power management, we are able to further optimize the electrical architecture. Different variants help tailor these to our customers individual electrification strategy and broader approach to architecture.

Next, Aptiv consolidates Input / Output which allows our customers to reap additional benefits from zone control, including simplifying vehicle manufacturing, and providing cost- and power-efficient compute for body function up-integration. As highlighted by Martin and Christian, this is a meaningful step which helps reduce the complexity and cost of their vehicles.

Finally, zonal data routing helps simplify sensor management to enhance the scalability and flexibility of advanced safety systems. Aptiv supports pre-processing, compression and routing of detection-level sensor data where required, while a premium variant will add fail operational performance for level 3 functionality and above.

Aptiv will be first to market with zone control with start of production in 2022, and we look forward to leveraging the insights from our unique position with both the brain and nervous system of the vehicle to tailor these solutions to our customers individual technology roadmaps and architecture strategies.

The move to smart power management is taking off, as evidenced by the dramatic acceleration in powertrain electrification we have seen over the last year. Let me hand it back to Glen to hear more.

ACCELERATING HIGH VOLTAGE ELECTRIFICATION

Glen De Vos | Senior Vice President and Chief Technology Officer

Thanks, Lee. Electrification is an integral part of the SVA story. In the race to electrify fleets, Aptiv has emerged as a partner of choice, capable of moving quickly to help our customers reduce time to market.

Our capabilities were on display in 2020, as we moved at unprecedented speed to support our customers globally with a comprehensive portfolio and a high degree of vertical integration. Now here to provide a couple of real-world examples is Josie Archer.

Josie Archer | Vice President, Sales Electrical Distribution Systems

Thank you, Glen. We know speed to market is critical, and Aptiv knows how to move fast.

For example, when a customer was experiencing issues with an incumbent supplier for an electric vehicle platform, Aptiv was able to offer three proposals with working hardware in less than one month: One that was designed by the OEM customer without changes; One that improved the design by leveraging Aptiv's extensive portfolio to substitute better and lower-cost components and add low-risk optimization enhancements; And a third that applied advanced innovations to maximize optimization of the overall weight, mass and cost.

Our final design took advantage of those capabilities by: Reducing the weight on heavy gauges cables by 40% through alternative material technology including aluminum and bus bars; Reducing the weight of the traditional gauge cables by 30% through optimization of routing and more sophisticated cable management; And we eliminated complex welded splice interfaces with Aptiv splice technology allowing for a 45% reduction in the size of the interfaces while enabling automation and increasing durability.

As such, we began shipping 8 months later – far faster than the typical three-year lead time from business award to start of production.

When another customer, a leading EV manufacturer in North America, wanted to expand production to China, and do it fast, Aptiv's global capabilities allowed us to go from business award to SOP within 4 months. Aptiv utilized global best practices, as well as common processes to quickly bring a solution to the customer in a familiar way.

Here are the factors that allow Aptiv to move with speed to get to market fast: 1) A comprehensive portfolio means not having to coordinate with a half dozen suppliers around the world. 2) Optimized system solutions means better options for balancing cost and performance. 3) A modular approach gives us the ability to offer the right components for the right application – whether that's choosing between busbars or round wires, or between different inlet modules for different regions. 4) Cable extrusion expertise means we can make cables in-house in all major regions, which reduces overhead and shortens the supply chain. 5) A common engineering and manufacturing footprint results in a more efficient development process and simplifies logistics for production and change management. 6) And finally, in-house rapid prototyping capabilities mean we can print production-grade components in real time to support concepts or advanced development programs.

Chris Reider | Vice President, Global Engineering - Connection Systems

To support the demanding requirements of electrification, we are excited to launch our family of High Voltage interconnects. This connector family has been purpose to support the higher power requirements associated with larger battery packs, with faster charging, and for the increasing demands of traction motor systems, while maintaining peak performance throughout the life of the vehicle.

Some innovative features include: Support for 400 amp duty cycles for faster charging and 300amps continuous for traction motor systems; It has a direct-contact terminal system with integrated stainless steel spring, that delivers a 100X life projection over traditional copper-alloy terminals; Its compact design enables reduced mass, simplifies packaging, at a reduced cost is available in a variety of configurations, including direct-contact blade design that simplifies connections and allows current to flow directly through high-conductivity copper.

These families are tooling up today, and validation will be completed by mid-2021.

ENHANCING THE IN-CABIN USER EXPERIENCE

Glen De Vos | Senior Vice President and Chief Technology Officer

Thanks, Chris. This is a great addition to Aptiv's comprehensive line of high-voltage components, and we look forward to making them a vital part of the next-generation of electric vehicles.

Now we've talked about a number of SVA principles today – from electrification, to the up-integration of compute with Aptiv Zone Controllers, and abstracting hardware from software in the Aptiv Next Generation ADAS platform just to name a few.

Many of these same advancements apply directly to the in-cabin user experience, and it comes at a pivotal time for the industry. Consumers are expecting a rich user experience from their vehicles – one that is more intelligent, integrated and connected. A seamless experience between the passengers, the vehicle and the world around it.

For vehicle infotainment systems to provide that experience, and to continue to bring value to consumers, they must overcome some very real challenges specific to automotive applications, including lifecycle management expectations, app availability, speed to market and security.

Aptiv has the architecture vision as well as the technical capabilities, to meet new challenges while collaborating with our customers to develop fantastic user experiences. We have deployed solutions across every major operating system – most notably Android – while tailoring the solutions to OEM and region-specific requirements.

Specific examples can be seen on Aptiv's current-generation systems, which are available on vehicles today. Volvo and the Polestar 2 is the first to market with native Android Automotive running Google Automotive Services. Our Android software and hardware platform allows full application abstraction, enabling our customer to deploy Google Services globally while seamlessly deploying a different regional ecosystem provider.

The Audi / Porsche MIB3 is one of the world's most complex and capable integrated cockpit controllers ever developed. It was first launched on the Taycan and is now rolling out on other vehicles.

And finally, Great Wall Motors' V3 ICC was the first Integrated Cockpit Controller to be deployed by a Chinese OEM. First launched on the Haval and Wey brands, this system implements two functional domains: an ASIL-B cluster, and an Android domain for infotainment that hosts Tencent's TAI app ecosystem.

Here's Alwin Bakkenes to show us some recent implementations of Aptiv's user experience solution, and to give us a glimpse into what's coming next.

Alwin Bakkenes | Vice President, Global Product Lines Advanced Safety & User Experience

At Aptiv, we strive to empower OEMs to control and enhance the software that defines their vehicles. This enables and infinite amount of individual experiences for their customers as they form new memories on the road ahead. At Aptiv, we believe that a more connected user experience starts with open platforms, deeply integrated with the sensing, compute and interface devices that support them.

As a result, our next generation platform takes a highly centralized and software defined approach, which represents our Smart Vehicle Architecture, and provides the most comprehensive in-cabin user experience platform available.

Up-integrated into the next generations platforms will be three domains: the first one is infotainment, then driver information and interior sensing - resulting in approximately 30% savings versus a standalone ECU while improving overall performance.

The infotainment domain is based on Android and fully compatible with Google Services as well as AOSP requirements. With full software abstraction, we can easily support a range of open platforms to meet OEM and regional requirements.

The driver information domain implements services such as rear-view camera, system sounds, and tell-tales, guaranteeing sub-second startup times from cold boot, while reducing cost through up-integration.

Interior sensing is evolving from basic driver monitoring for safety to more sophisticated comfort and convenience functionality, so up-integration into the platform is a natural evolution here too. We're implementing a full suite of applications, including driver and cabin sensing solutions, on a separate domain abstracted by an interior sensing feature framework.

Now with multiple functional domains, automotive software IP, and life cycle services, Aptiv offers a complete product platform for developing best-in-class in-cabin user experiences while reducing the cost of ownership .

CONCLUSION

Glen De Vos | Senior Vice President and Chief Technology Officer

We hope you've enjoyed this brief but compelling look across Aptiv's exciting suite of next-generation products, platforms, software and solutions. While 2020 proved Aptiv could deliver for our customers despite significant industry disruption, we also continued to innovate at a record pace, bringing to market exciting new technology solutions that continue to help solve our industry's toughest challenges. That speed and agility is what defines us, as we move with purpose to put Innovation in Motion!

Our capabilities in vehicle architecture, high voltage electrification, advanced safety and user experience, and connectivity bring together our unique portfolio of brain and nervous system capabilities. These full stack solutions enable our vision of Smart Vehicle Architecture, providing a roadmap to customers to advance the feature functionality consumers are demanding, and do that in a more cost-effective manner compared to any other alternative.

As we look to this year and beyond, Aptiv will continue innovate in new ways. Each Innovation in Motion will further expand our competitive moat and take us closer to our mission of delivering a safer, greener and more connected future of mobility. We look forward to taking your questions in our live follow-up Q&A session. Thank you!

LIVE Q&A SESSION

Aptiv Speakers:

- *Glen De Vos | Senior Vice President and Chief Technology Officer*
- *Elena Rosman | Vice President, Investor Relations*

Elena Rosman: Welcome back – we are now going to start the live Q&A portion of today's event. As a reminder, to ask a live question, please use the dial-in details on the event website below this live stream. Our conference operator will now provide additional instructions.

Conference Operator: Thank you. If you would like to ask a question, please signal by pressing *1 on your telephone keypad. If you are using a speakerphone please make sure your mute function is turned off to allow your signal to reach our equipment. Also please make sure to mute your live stream before asking a question.

Elena Rosman: Thank you, Tracey. While we are assembling the queue, I will kick it off with the first question…

Glen – I'll start with one on SVA: During Aptiv's 2020 CES event, featuring SVA™ you highlighted how SVA creates lower cost of total ownership savings upon full deployment. Has the work you and the team have done with our advanced development program partners, especially in the areas of zone control, validated the value proposition to the customer and what are the next milestones to gauge customer interest?

Glen De Vos: Thanks Elena. A year ago, it was kind of the plan in the anticipated savings for SVA what we thought we could deliver. Over the past year, we've really focused with our customers on implementing that and seeing and validating those savings and the value prop thesis.

What's been great is it's really held. And we've seen that by up-integrating compute, we're able to drive 25% lower total cost with content and compute. We've seen a significant reduction in the weight of the wiring harnesses.. so simplification and reduction of complexity in the wiring harnesses. All of which really supports the fact that with the zonal architectures and moving in that direction, you lower that total on cost of the vehicle.

In addition, we're really saying that with a SVA approach, you simplify the deployment of these systems and have significant savings in terms of the implementation. We talked about 75% less costs associated with implementing testing and validating these systems, so that's been really good to see. As we think about the year that we've just had, it's been a really strong proof point for us.

So as we kind of pivot going forward now, what do we expect to see next? Over the course of 2021, that'll be when we more or less see the plan to go to market or the adoption and the actual implementation plans for zonal controls for further centralization of compute. And we expect -- even though we're launching our initial zone controllers in 2022 -- for that to take place in the 2024 timeframe. That's what the announcements have been about here today: our zonal architecture.. looking at the zonal controllers, as well as our next generation ADAS platform, which is the further centralization of compute, and continuing with that theme of SVA. So we're looking forward to an equally exciting year ahead.

Elena Rosman: Thanks, Glen. Operator, can you open up the line for questions?

Conference Operator: We will now take our first question from Rob Lache from Wolfe Research.

Rob Lache | Wolfe Research: Regarding deployment of localized zonal control in terms of penetration maybe in 2025, or 2030. You did say that the first applications are coming already now, or in 2022, and expanding in 2024. But can you give us a sense of how broadly this is going to be deployed within OEMs?

Is the organizational structure of your customers still kind of an impediment? They are organized in sort of a siloed way with siloed functionality? What's the competitive landscape?

And can you talk a little bit about the tradeoff between optimizing wiring and the gains that you get in terms of the optimized architectures?

Glen De Vos: Sure. So let me let me start with the timing aspect. In 2024/2025, we'll be seeing the initial application. Elena, may be able to add some more color on exactly how we view the market. But that's again, where we would expect to see that coming in.

Relative to competitive landscape, we're uniquely positioned here in the sense that we have both compute what we think of as the Brain, as well as the Nervous System or the electrical architecture piece. We see this as a great opportunity for Aptiv because by sitting down and working with those OEMs, we're able to optimize, look at, and work through those tradeoffs of centralized compute versus distributed, as well as how to simplify the electrical architecture. And that's really been the process that's worked well over the past year.

As it relates to the savings associated with that.. when we more or less partitioned the vehicle into zones up integrated the compute within the zones, what we saw there was approximately nine different modules being consolidated into one centralized computer and power data center, as we call it. And effectively that's what resulted in about the 25% weight, as well as packaging, volume reduction, which tends to translate to system cost, to the wiring.

Through the simplification that the zonal architecture brings, we're able to dramatically reduce the number of interconnects, between boxes and overall wiring length -- what that allows for is a light weighting of those systems. We took at about eight kilograms out of that out of that system, and then reduced the wiring by 20%, in terms of total weight for those zones. So I think as we looked at it, it validated the overall thesis that by going zonal, you'll significantly reduce complexity and cost for the OEM's total cost.

So, maybe Elena if you want to comment on that?

Elena Rosman: Yes. So the evolution is exactly playing out as we predicted and really starts with our first launch of zone control in 2022. We'll see continued scaling of that really in that 2025 and beyond timeframe. We believe the market will grow pretty rapidly, but somewhere between 15 and 20 billion before we see 2030.

So I think it's obviously an area that we feel like we will achieve high market share, significantly above where we have market share today in and around body and ECUs. We really benefit from being at that point of content aggregation, as we see more highly integrated centralized compute going into the vehicle.

Glen De Vos: And Rob, you asked one other question regarding the OEM org structure, with the zonal. In that area and those domains, what you're really talking about is the body domain, the electrical architecture domain. Within the OEMs, those are fairly close in there. They're adjacent and quite frankly, work really well together.

If you think about body and what's involved there, it's very impactful to the electrical architecture. And as such, we really don't see any organizational barriers per se. In fact, as we've worked with those OEMs, that's been really an exciting part of it -- as they recognize that there is a real opportunity here that they can make a significant improvement on the overall vehicle architecture. Those organizations work very effectively together and so that's what's helped us really kind of accelerate.

Rob Lache | Wolfe Research: And you mentioned that there are new business models and revenue opportunities for Aptiv. Related to this, how do you see that playing out? What kind of incremental opportunity do you expect? Within this visible timeframe that you see in 2022 and 2024.

Glen De Vos: Yeah, there's really two aspects of that. One is by being the provider of the zonal controller, your point of up-integration. We think that's the content per vehicle dimension to it, where -- similar to what we've done with other domain controllers -- that point of centralization. There’s an interesting opportunity for us there.

Along with that with zonal controllers comes the abstraction of software from hardware, and that unlocks a couple of different business opportunities for us. We think of that in three areas: One is the, if you think about being that systems integrator or that point of up integration, there's engineering services that are part of performing that role where you're integrating and constantly updating and then supporting the product. So there's an engineering services play related to the software.

The second is, and we're seeing this now starting, the selling of IP is kind of a pure play software IP sale. And so that's selling licensing features into the controller itself. And as I mentioned, that's really just starting now but it's enabled by the fact that the software is separated from the hardware, and you can sell discrete modules of code or financial functions and features to the OEM to be applied. And then as a systems integrator, we would integrate those modules. That's beginning now so we're starting to see that happen.

Then the third really relates to lifecycle management of the product. That's the ongoing support that's required for essentially maintaining the product over its life. If you think about product today, the vast majority is software embedded in hardware. Now we're talking about software extracted with OTA capabilities, so we're setting up support for managing that software over the lifecycle of that vehicle, and actually beyond into other applications.

Generally speaking, those are the types of opportunities that are really opened up. If you think about today, hardware and software embedded and you break that apart, we would see 80% of that still really is that hardware sale with the embedded software component. But about 20% of that really is represented by these other revenue opportunities, and obviously with the opportunity for expanded margin.

Conference Operator: We will now take our next question from Dan Levy from Credit Suisse. Please go ahead.

Dan Levy |Credit Suisse: Hi, good morning. Thank you, very helpful and appreciated. First, one to ask can you just remind us in practice, how SVA will actually be adopted. And what I mean by that is we've seen automakers unveil their own electrical architectures, GM like as they unveil new EV platforms. So what pieces of SVA are actually adopted by the automaker as opposed to being done in house?

Glen De Vos: Yeah, so I think there's two questions in there. One is, what's the roadmap for SVA, how do we see that playing out? And then the other is, well, how does that balance between what the automaker would feel like they want to do themselves versus what an Aptiv would do?

So with regard to the first…SVA as you think about it, it starts with domain centralization. This actually started some years ago, and really with the Audi zFAS launch back in 2017. That was the first real example of domain centralization, in this case the ADAS domain, and that continues.

Now what's happening is what we just talked about zonal architectures. That's really where you're breaking the, you can think of it as the edge of the car. It's everything that's distributed around the vehicle sensors, actuation power distribution -- breaking that piece of the car into zones. That's what zonal architecture is all about and that's what's starting now.

You're hearing a lot more about the OEMs recognizing the importance of doing that, because it reduces the complexity of the wiring harnesses. We can't just continue to add wires and copper to the vehicle, so it simplifies the electrical architecture. It provides an opportunity for up integration of all those many different little modules around the vehicle, the ECUs into what we call zonal controllers. That's the real kind of second step and that's happening now.

And what the other important part of that second step is, that's how you now manage and basically separate IO, or all the IO from those sensors, and all those different components, going back to the central compute, the centralized controls -- that's step two.

Step three is really then the continuation of that domain centralization into really what we call serverization of compute. If you think about it, you now have your IO in a ring topology around your compute. That's what enables you to then serverize or take that next big step of fully centralizing that compute into could be 1, 2, 3, or 4, high performance compute that supports all of the ADAS, the propulsion, the infotainment, the connectivity domains.

That's what we see is really happening and starting to happen broadly in about the 2025 timeframe. That's really what we consider to be kind of that roadmap to SVA.

Now, in terms of what are the OEMs do versus the suppliers, like an Aptiv or the partners like Aptiv, that varies by OEM. It starts with the fact that the OEMs do need to and want to have control over the software in the vehicle. You see different strategies being rolled out across the OEMs of how much of that they want to control. In general, there's a theme that the OEMs want to be able to see and be able to control the software that's going into their vehicle, and then really manage it over the life of that vehicle.

Where exactly you draw the line between what the OEMs do and what the partners like an Aptiv do, it really varies. In some cases, we provide turnkey systems with full feature content, as well as all the base software and the ongoing lifecycle services. In other cases, we provide a platform on which the OEMs can innovate and put their content.

The importance for us is that we remain and have a flexible model, a working model that can really tailor to those specific OEM strategies. We know we don't dictate to an OEM or have a blackbox approach. We have an approach where they can innovate, and they can have the content that they want to control as part of that platform. Or they can take it as a turnkey system from Aptiv. That's been, I think one of the key successes for us in our in our satellite architectures that we're launching now. It's really been an important factor that we can be really flexible with the OEMs work models.

Conference Operator: We will now take our next question from Ryan Brinkman, from JP Morgan. Please go ahead.

Ryan Brinkman | J.P. Morgan: One related to Smart Vehicle Architecture and other related to next generation ADAS. Can you talk about the degree to which your expertise in electronic architecture helps you to better understand or go to market with customers with ADAS solutions? Or vice versa?

Does your having both the signal and power distribution and the advanced a few user experiences help businesses under one group? Does that provide you with any competitive advantages as both of the businesses undergo transitions to the next generation of these systems? Do your competitors similarly combine any such capabilities? Thank you.

Glen De Vos: Yeah, it's a great question. There's no doubt that understanding the electrical architecture and the physical and data transmission requirements of connecting all these devices is an important part of our value prop. That was one of the reasons we went to the satellite architecture to begin with.

When we looked at just the physical interconnection and then how you manage data around a bunch of smart connector or smart sensors that are trying to perform ADAS functions, one of the things we recognized was it's much more efficient and much better performing to use a satellite system.

Simplifying the sensors, bringing that back to a central controller, doing that either over Ethernet or high speed data interconnect that our SPS team brings to us...that knowledge and understanding was a big part of why we recognized early on that the move towards more centralized compute was the right move.

As you think about it on a go forward basis, that just continues because as more compute, more software, more data, more sensors are coming into the car, the challenge that creates for the electrical architecture just increases. It's really been that SPS piece, the electrical architecture understanding, that has been so instrumental in forming our strategies around smart vehicle architecture, and recognizing that we simply couldn't continue down the current path of distributed architectures with smart sensors and all of the interconnect that that was dragging with it.

As we think about that next gen architecture, we're building on that understanding that we get from our Signal and Power Solutions Group about what does it mean from a physical interconnection and a data management standpoint. That really extends not just to the ADAS Gen II piece, but as part of zonal architecture. It's part of our infotainment strategy. It really hits us in all of those different areas. We really leverage that capability, and it's been a great part of what we can bring to our customers.

When we think about our traditional competitors in the compute or the ADAS side, they really don't have that capability. They have to bring in an outside supplier or work that through the OEM. Having that in house, where those engineers sit side by side, we feel enables us to come in with more efficient solutions, and focus on lowering that total cost of ownership for our customers.

Conference Operator: We will now take our next question from Steven Fox from Fox Advisors. Please go ahead.

Steven Fox | Fox Advisors: Hi, good morning. Thanks for taking my question. Just one from me, please. Could you just talk a little bit about how the Smart Vehicle Architecture and some of your other initiatives affects semiconductor procurement?

In some ways, you seem to be driving some higher end workloads. In other ways, it seems like you may or are able to optimize semiconductors so that the total purchasing price or the focus on the high end isn't so big. Can you talk about how you're helping your customers do that? Thanks.

Glen De Vos: Sure, there's two benefits. The first is, as we think about centralized compute and as you were saying Steve, it does drive to more higher performance microprocessors in those compute modules. So as you do that, you have to have the performance you need there. But the way we've architected the platforms, whether it's zonal control, the next generation ADAS or its infotainment, we do this in a way that really is abstracted from the underlying compute and not dependent on specific semiconductor architectures. Whether it's x86 or its ARM, we try to maintain independence from them.

What that allows you to do is take advantage of the best processing capability that's out there and not be as restrictive in terms of what silicon you can use. We think that's a really important piece of the overall strategy in terms of making sure you have not just the best available, but you can work with a variety of semiconductor providers, which we think is always smart.

As you think longer term towards more centralized or the serverization of compute, we're fundamentally procuring compute, not unlike what you do with servers. That means you buy compute based on performance -- the spec KPIs or the performance needs of that compute.

That means you have a lot more flexibility in how you deliver that compute. You're not going to be locked into specific architectures. You're really looking at it from an overall performance standpoint, and buying compute in terms of capacity, as opposed to specific silicon architectures.

Again, we think that's an important piece of this strategy, because it allows you to take advantage of the different compute solutions that are out there that are being provided by the industry. It doesn't lock you into, a narrows silicon solution. Our goal is to make sure that the industry/ our customers can take advantage of the most advanced compute, but not be locked into very narrowly defined solutions.

Conference Operator: We will now take our next question from Joseph Spak from RBC Capital Markets. Please go ahead.

Joseph Spak | RBC Capital Markets: Thanks very much, miss the Vegas experience. Thanks for thanks for doing this. If we think about SVA, and I know you're in varying stages of progressing development trials with the traditional customers, meaning the automakers, but it also seems like there are companies that are looking to provide a platform for third parties and you know, electric vehicle platforms. If you view SVA as having the opportunity to help provide the full stack solution for some of those platforms that is being developed for third parties?

Glen De Vos: That's a great question Joe, because I think what you're seeing with some of those developments is a move towards essentially more stable architectures especially with regard to BEV approaches. There's a kind of an alignment that's occurring within the industry. It's very much one of the things that we've seen over this past year since we talked last January. It has been that the trend towards electrification is bringing with it, this acceleration towards SVA and these more advanced architectures that are simpler, more efficient. They're very much lining up with the key elements that we've been talking about, which is the centralization of compute, the move towards zones and simplification of the architecture around the edges.

For Aptiv, where our approach aligns well with a with a customer's offering, there's a great opportunity for us. I would say, both on the high voltage as well as the low voltage side, and then and that's been probably one of the real exciting parts about 2020.

Going into 2021, we see that working with some of the new entrants into electrification. This approach, SVA, is very much aligned with how they're thinking about architectures. It gets back to the fact that when you go electric full BEV, you have an opportunity, with more or less a kind of a clean sheet on the architecture. You can really step back and look at it and approach it as a software defined platform, and then build it out from there.

That's exactly what SVA is trying to do. So for us, it's been great working with some of the new entrants and talking through how we can help support them both on both the low voltage as well as high voltage.

Conference Operator: We will now take our next question from Colin Rusch from Oppenheimer, please go ahead.

Kristen Owen | Oppenheimer: Hi, this is Kristen going on for Colin, thank you so much for taking our question. I'm just wanting to pick up on the speed to market from award to production. Are you seeing any dramatic changes in vehicle development cycles or in shortening of that time from award to production at this stage?

Glen De Vos: I'm not sure, I would say not dramatic. The overall vehicle development lead times have been shrinking over the past years, nothing dramatic yet.

Now, what I would tell you is though, as we go to more abstracted software and more advanced vehicle architectures, I think that sets the stage for faster design cycles -- as well as managing, the development of particularly software content for vehicles in a very different way.

What I mean by that is, if you think about today's environment, there's a ton of work that goes into and right up to the launch of that vehicle…getting everything ready, getting all the software bugs fixed. I think in 2020, we saw some examples of how OEMs, were struggling with that complexity and getting that all put together and working.

As you think about a software defined vehicle where you're doing continuous integration/continuous deployment of that software, you're now managing that software essentially independent of that underlying hardware platform of that vehicle. And as those vehicles are developed, and they come on, you then essentially port the software over to the vehicle that allows you to be much more efficient in terms of that system development.

That's why we point out that there's a significant savings associated with SVA's approach to vehicle architecture. It enables you to manage that software complexity somewhat independently. As you think about going forward with electrification in particular, those are platforms that lend themselves and those are architectures that help accelerate time to market and vehicle development.

I think you're going to see that now kind of accelerate, so I would expect shorter vehicle development cycles going forward. Today, not really big, but I think we're right at that inflection point where you're now changing how you manage the software content, the vehicle, the architectures, electrification, all of those things help with reducing vehicle lead times.

Conference Operator: We will now take our next question from Itay Michaeli, from Citi. Please go ahead.

Itay Michaeli | Citi: Great, thank you. Good morning, and thanks for hosting the event. Just a couple of questions on ADAS Glen. On the imaging radar, the FLR4 I believe is that still entering production in 2023? And how do we think about the CPV of imaging radars in general relative to what you're selling today?

Glen De Vos: Yeah, so the imaging radar is FLR4. That's really that next generation and '23 is about the right timeframe for that. And what that brings is much greater distance, the inclusion of machine learning and AI techniques in terms of object discrimination and classification. That's really that next big step. That's part of our next generation ADAS platform because it really opens up more use cases and more capability for the system.

I'll let Elena talk about the revenue side of the ASP side. But, our goal with that is to be able to utilize the more advanced radar to actually still simplify the total system. So you can avoid having either as many radar or in fact, include other sensors or reduce the complexity of those sensors because you have more capability in that radar. Maybe I'll let you speak to the CPV side of it.

Elena Rosman: We don't typically give our CPV s for individual components. We've talked about the content per vehicle opportunity within ADAS. As you move from Level 0 / Level 1applications to Level 2/ Level 2+ and there on.. so as a reminder, the content per vehicle opportunity for Aptiv for a Level 0 / Level 1 application is really in that range of call it $250. That grows, and you get to a Level 2 system to an upwards of $500 per vehicle. A Level 2+ system again - that includes now multiple radars, including the side radars, the forward looking radars - would be an excess of $750 for a Level 2 application. When we really get to Level 3 and the inclusion of some LIDAR, in addition to bigger compute more software capabilities, that's an upwards of $3,000- 4,000 of content per vehicle.

Conference Operator: We will now take our next question from Brian Johnson from Barclays, please go ahead.

Brian Johnson | Barclays: Thank you. In terms of translating this to revenue growth for in particular for the SPS segment just based on your discussions with OEMs, and also in light of the earlier question about which OEMs are going to buy the whole thing versus parts versus license technology? When should we think about this meaningfully inflecting revenue growth?

Glen De Vos: So Elena, maybe I can just talk a little bit about the timing that we've laid out, and then you can speak to the revenue piece of it. So Brian, great question. As I mentioned, what we're seeing right now with what's particularly impactful for SPS is that move towards zonal which occurs in that '24/'25 kind of timeframe. That's when we would expect that to really hit so maybe Elena, you can speak more about the revenue side of the equation.

Elena Rosman: So again, our first launch on zone control is in 2022. We will expect zonal control, that's, that's additive from a content per vehicle, and from a market share perspective, really starting to kick in that 2023 timeframe scale, in 2025 and beyond. So we see that revenue going into our connectivity and security product line within the Advanced Safety and User Experience segment. So we would expect to see, you know, some wins in 2021. And that sourcing to really begin in earnest over the next you know, call it 12 to 18 months to support that next, that next step towards Smart Vehicle Architecture with zone control.

Conference Operator: We will now take our final question from David Kelly from Jefferies. Please go ahead.

David Kelly | Jefferies: Hi, good morning Glen and Elena and thanks for squeezing me in. Maybe if we could just follow up on the earlier radar discussion, Glen? Is it fair to say that you see radar taking on an increased load as it relates to the roles the various sensors are going to play in object detection going forward?

And then secondarily, does that in your mind reduce the total system costs for your customer ultimately? Just trying to get a sense for if this could accelerate rollout.

And then finally, if I could squeeze it in potential timeline of that level three pro platform? Just would love to get a sense for how you all are thinking about that rollout as well.

Glen De Vos: Sure. We've always been more or less a radar centric platform. If you think about our first generation platform, it operates at the Level 2 with five radar - a forward looking and four corner radar - as well as a forward looking camera system that really provides that full functionality.

Now, as we go to our next generation of imaging radar, what that does for us is it allows us to essentially use radar to do more of the heavy lifting by taking advantage of what it does well -- range, relative velocities, all weather types of performance, etc. Those radar have a lot more capability, a lot more technology going into them. But at the same time, I can now with the four radar do what I previously was doing with five. I can use those imaging radar towards the front corners, and that actually covers and provides the coverage in the forward looking direction as well. That gives you some really interesting options in terms of reducing total system complexity and cost, while delivering greater levels of performance.

Another aspect is, to the extent that I can leverage radar more, especially in like a Level 2+ or Level 3 application where I might be needing a LIDAR, I can potentially eliminate the need for that in my Level 2+, certainly at lower speeds for like traffic jam assist or kind of urban applications.

So now I'm able to offer a L3 system at the lower speed and lower velocity, and that doesn't require a LIDAR. That gives me a lot more options in terms of bringing more functionality to the customer at a lower total cost. It also reduces the burden on the vision systems. I don't have to go to as high megapixel camera or as many cameras to be able to deliver the functionality, all of which reduces camera costs, as well as processing costs.

As we think about that next generation architecture, the really kind of a great aspect of it is I can configure it in many different ways. As a satellite architecture, I can configure those sensors in a variety of different setups that give the OEM the functionality they want at the lowest possible price point. That gives us a lot of what we call flexibility and scalability.

In terms of timeline for Level3, there still are some introductions that are planned to occur this year. I think those will be relatively limited volume –limited what we would call operational design domain or functionality. This is very specific use cases. Others are really talking of rolling out in 2022 to 2024 timeframe. As we think about our next gen system that would incorporate this capability that lines up nicely with that in the 2024 timeline.

Elena Rosman: Glen, I just wanted to add.. Given that our solutions are data and processing, we have seen about a 30% cost improvement versus prior versions of the radar. As Glen mentioned, it certainly further reduces some of the dependency on a vision for Level 2/Level 2+, and obviates some of the need for some of the expensive LIDAR in L3 applications.

These savings are affording higher levels of ADAS features as OEMs are looking to democratize their active safety across their full vehicle lineups. We've seen that as part of our rollout of our satellite architecture, and really upgrade their capabilities to Level 2 / Level 2+, which includes five total radar systems.

Glen De Vos: Yeah, and I think Elena, you really make an important point when we think about SVA, or we think about the zonal controllers and the ADAS architectures that we're rolling out today…It really is about how do we democratize those technologies? How do we lower that total system cost for the OEMs so they can add these features in without adding a lot of costs and complexity to the vehicle.

Elena Rosman: Thanks, Glen. Thank you all for joining today, thank you for your active participation during our Q&A session! This now concludes today's event. As always please feel free to contact me or IR team. If you have any additional questions, we'll be available throughout the day and you know how to reach us. Thank you again for your participation. Have a great day!
